FIN2020 HW6 文杰 123090612

The aggregate endowment is the same in both states:

$$e^A + e^B = (1,2) + (2,1) = (3,3).$$

So there is **no aggregate risk**, only redistribution across agents. In an Arrow-Debreu economy with concave utility, trade will therefore fully smooth consumption across states in any Pareto-efficient allocation.

A. Characterize the set of Pareto optimal allocations

Consider the social planner problem with weight $0 < \alpha < 1$ on agent A and $1 - \alpha$ on agent B:

$$\max_{c^A, c^B} \ \alpha \left[\frac{1}{2}\log c_1^A + \frac{1}{2}\log c_2^A\right] + (1-\alpha)\left[\frac{1}{2}\log c_1^B + \frac{1}{2}\log c_2^B\right]$$

subject to the resource constraints in each state:

$$c_1^A + c_1^B = 3, \qquad c_2^A + c_2^B = 3.$$

Because the problem is separable across states, for each state $s = 1, 2$, the problem is

$$\max_{c_s^A, c_s^B} \ \alpha \log c_s^A + (1-\alpha)\log c_s^B \quad \text{s.t.} \quad c_s^A + c_s^B = 3.$$

Substituting $c_s^B = 3 - c_s^A$ yields:

$$\max_{c_s^A} \ \alpha \log c_s^A + (1-\alpha)\log(3 - c_s^A).$$

First-Order Condition (FOC):

$$\frac{\alpha}{c_s^A} - \frac{1-\alpha}{3 - c_s^A} = 0.$$

So

$$\alpha(3 - c_s^A) = (1-\alpha)c_s^A$$

$$3\alpha = c_s^A.$$

Hence

$$c_s^A = 3\alpha, \qquad c_s^B = 3(1-\alpha), \qquad s = 1, 2.$$

Therefore, the set of Pareto-optimal allocations is

$$\boxed{c^A = (3\alpha, 3\alpha), \qquad c^B = (3(1-\alpha), 3(1-\alpha)), \qquad 0 < \alpha < 1.}$$

So every Pareto optimum features **full insurance across states** for each agent.

B(a). Agent A's problem and security demand

Let p be the price of the Arrow-Debreu claim for state 1, and normalize the price of the state-2 claim to 1.

Agent A chooses state-contingent consumption (c_1^A, c_2^A) to solve

$$\max_{c_1^A, c_2^A} \ \frac{1}{2}\log c_1^A + \frac{1}{2}\log c_2^A$$

subject to

$$pc_1^A + c_2^A \le p \cdot 1 + 2.$$

Since utility is strictly increasing, the budget constraint binds:

$$pc_1^A + c_2^A = p + 2.$$

FOCs:

$$\frac{1}{2c_1^A} = \lambda p, \qquad \frac{1}{2c_2^A} = \lambda.$$

Dividing the two equations yields:

$$\frac{1/c_1^A}{1/c_2^A} = p \quad \Rightarrow \quad c_2^A = pc_1^A.$$

Substituting into the budget constraint:

$$pc_1^A + pc_1^A = p + 2$$

$$2pc_1^A = p + 2$$

$$\boxed{c_1^A(p) = \frac{p+2}{2p}}, \qquad \boxed{c_2^A(p) = \frac{p+2}{2}}.$$

Expressing **security demand** as net purchases relative to endowment gives:

$$x_1^A = c_1^A - 1 = \frac{p+2}{2p} - 1 = \frac{2-p}{2p},$$

$$x_2^A = c_2^A - 2 = \frac{p+2}{2} - 2 = \frac{p-2}{2}.$$

So Agent A's net demands are

$$\boxed{x^A(p) = \left(\frac{2-p}{2p}, \frac{p-2}{2}\right).}$$

B(b). Agent B's problem and security demand

Agent B solves

$$\max_{c_1^B, c_2^B} \frac{1}{2}\log c_1^B + \frac{1}{2}\log c_2^B$$

subject to

$$pc_1^B + c_2^B \leq 2p + 1.$$

Again the budget constraint binds:

$$pc_1^B + c_2^B = 2p + 1.$$

FOCs:

$$\frac{1}{2c_1^B} = \lambda p, \qquad \frac{1}{2c_2^B} = \lambda$$

which imply

$$c_2^B = pc_1^B.$$

Substituting into the budget constraint:

$$pc_1^B + pc_1^B = 2p + 1$$

$$2pc_1^B = 2p + 1$$

$$\boxed{c_1^B(p) = \frac{2p+1}{2p}}, \qquad \boxed{c_2^B(p) = \frac{2p+1}{2}}.$$

Net security demands:

$$x_1^B = c_1^B - 2 = \frac{2p+1}{2p} - 2 = \frac{1-2p}{2p},$$

$$x_2^B = c_2^B - 1 = \frac{2p+1}{2} - 1 = \frac{2p-1}{2}.$$

Thus

$$\boxed{x^B(p) = \left(\frac{1-2p}{2p},\ \frac{2p-1}{2}\right).}$$

B(c). Equilibrium price and allocation

Using market clearing conditions, since the total endowment in each state is 3, it must hold that:

$$c_1^A + c_1^B = 3.$$

Substituting the demands gives:

$$\frac{p+2}{2p} + \frac{2p+1}{2p} = 3.$$

So

$$\frac{3p+3}{2p} = 3$$

$$3p + 3 = 6p$$

$$3 = 3p$$

$$\boxed{p^* = 1.}$$

Now compute equilibrium consumption:

For agent A:

$$c_1^A(1) = \frac{1+2}{2} = \frac{3}{2}, \qquad c_2^A(1) = \frac{1+2}{2} = \frac{3}{2}.$$

For agent B:

$$c_1^B(1) = \frac{2+1}{2} = \frac{3}{2}, \qquad c_2^B(1) = \frac{2+1}{2} = \frac{3}{2}.$$

Hence the equilibrium allocation is

$$\boxed{c^{A*} = \left(\frac{3}{2}, \frac{3}{2}\right), \qquad c^{B*} = \left(\frac{3}{2}, \frac{3}{2}\right).}$$

Equivalently, each agent fully insures and consumes the same amount in both states.

B(d). Is the allocation Pareto optimal?

Yes.

From Part A, every Pareto-optimal allocation has the form

$$c^A = (3\alpha, 3\alpha), \qquad c^B = (3(1-\alpha), 3(1-\alpha)).$$

The equilibrium allocation is

$$c^{A*} = c^{B*} = \left(\frac{3}{2}, \frac{3}{2}\right),$$

which corresponds to

$$3\alpha = \frac{3}{2} \quad \Rightarrow \quad \alpha = \frac{1}{2}.$$

So the competitive equilibrium allocation is one of the Pareto-optimal allocations:

$$\boxed{\text{Yes, the equilibrium allocation is Pareto optimal.}}$$

This is consistent with the First Welfare Theorem: under complete, competitive markets, competitive equilibrium allocations are Pareto efficient.